

Reserves Project

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Introduction

Reserves refer to the funds or resources set aside by organizations or governments to cover future obligations, uncertainties, or emergencies. These can include financial reserves like cash or securities, as well as natural reserves such as oil, gas, or minerals. In banking, reserves often refer to the capital that banks are required to hold to ensure they can meet their obligations. Essentially, reserves act as a safety net to ensure stability and preparedness for unforeseen events.



The main aim of our project is to analyze and understand the distribution, trends, and factors influencing reserves across various countries using statistical methods and visualizations. Major objectives of our project are stated as follows:

1. To explore the patterns and trends in reserves (e.g., foreign exchange reserves, gold reserves) across countries or regions using statistical tools in R.
2. To examine how reserves vary geographically and temporally i.e. from region to region and from time to time.
3. To identify potential relationships between reserves and other economic indicators (e.g., GDP, population, trade balance).

Importance and Types of Reserves

Importance

Reserves are critical for economic stability, facilitating trade, and managing financial crises. They help stabilize currency exchange rates, control inflation, and ensure smooth international transactions. Reserves act as a buffer against economic shocks, support debt repayment, and boost investor confidence, enhancing a country's creditworthiness and financial resilience.

Types of Reserves

Reserves come in various forms depending on the context in which they are used. Here are some of the major types:

Financial Reserves

1. **Capital Reserves:** Funds set aside from profits for specific purposes such as future investments or debt repayment.
2. **Revenue Reserves:** Funds retained from earnings to cover future expenses or emergencies, ensuring smooth business operations.



Natural Reserves

1. **Oil Reserves:** Quantities of crude oil that are known and recoverable under existing economic and operating conditions.
2. **Mineral Reserves:** Known quantities of minerals that can be economically extracted and processed.

Central Bank Reserves

1. **Foreign Exchange Reserves:** Assets held by a central bank in foreign currencies, used to back liabilities and influence monetary policy.
2. **Gold Reserves:** Quantities of gold held by a central bank as part of its financial assets.

Banking Reserves

1. **Required Reserves:** Minimum amounts of funds that a bank must hold in reserve against deposits, as mandated by the central bank.
2. **Excess Reserves:** Funds held by banks that exceed the minimum reserve requirements, providing additional liquidity and security.

Ecological and Environmental Reserves

1. **Biosphere Reserves:** Protected areas that aim to conserve biodiversity and promote sustainable development.
2. **Marine Reserves:** Specific ocean areas where human activity is restricted to protect marine ecosystems and biodiversity.



Each type of reserve plays a critical role in its respective area, providing stability, security, and resources for future needs.

Source and Key Variables

Source of the Data

The data for this project is sourced from the [World Bank](#), a reliable global database that provides comprehensive statistics on reserves and other economic indicators for various countries.

Key Variables Description

1. **Country:** Name of the country being analyzed.

Type: Categorical. Purpose: To compare reserves geographically.



2. **Year:** The year for which the data is recorded.

Type: Numerical. Purpose: To analyze trends over time.

3. **Reserves (USD):** Total reserves held by a country, measured in U.S. dollars.

Type: Numerical. Purpose: The main variable of interest, representing economic strength.

4. **Region:** The region or continent where the country is located.

Type: Categorical. Purpose: For regional analysis of reserves distribution.

5. **GDP:** Gross Domestic Product of the country in the same year.

Type: Numerical. Purpose: To explore correlations between reserves and economic size.

6. **Population:** The total population of the country.

Type: Numerical. Purpose: To analyze reserves per capita.



These variables will help in understanding the distribution, trends, and relationships associated with reserves.

Exploratory Data Analysis

Exploratory Data Analysis (EDA) is an essential step in the data analysis process. It involves summarizing and visualizing data to uncover patterns, spot anomalies, test hypotheses, and check assumptions using statistical graphics and descriptive statistics. EDA helps analysts understand the data's structure, identify outliers and missing values, and generate insights that inform further analysis or decision-making. It involves:

- **Summarization:** Using descriptive statistics like mean, median, and standard deviation.
- **Visualization:** Creating histograms, box plots, scatter plots, and bar charts.
- **Data Cleaning:** Handling missing values and outliers.
- **Pattern Recognition:** Identifying trends and correlations.

```
options(warn=-1)

file_1 <- read.csv("C:\\Users\\Abhinav Masih\\OneDrive\\Documents\\One Drive Folder\\OneDrive\\Desktop\\RESERVES FILES\\Bank
Reserves.csv")

# summarization using descriptive statistics
summary(file_1)
```

```
##      Country      Year  Cost.to.export  Cost.of.Import
## Length:234      Min.   :2014      Min.   : 415      Min.   : 410
## Class :character 1st Qu.:2014      1st Qu.: 1026      1st Qu.: 911
## Mode  :character Median :2014      Median : 1482      Median :1283
##                      Mean  :2014      Mean  : 1885      Mean  :1566
##                      3rd Qu.:2014      3rd Qu.: 2174      3rd Qu.:1807
##                      Max.   :2014      Max.   :10650      Max.   :9050
##      Reserves      Description      Magnitude
## Min.   :-7.491e+07 Length:234      Length:234
## 1st Qu.: 6.313e+07 Class :character Class :character
## Median : 8.607e+08 Mode  :character Mode  :character
## Mean   : 2.000e+10
## 3rd Qu.: 5.474e+09
## Max.   : 6.520e+11
```

```
mean(file_1$Reserves)
```

```
## [1] 20001685717
```

```
median(file_1$Cost.to.export)
```

```
## [1] 1482.25
```

```
sd(file_1$Cost.of.Import)
```

```
## [1] 1118.164
```

```
# Correlation between variables
cor(file_1$Reserves, file_1$Cost.to.export)
```

```
## [1] -0.06377386
```

```
# data cleaning
file_1 <- na.omit(file_1) # handling missing values
file_1 <- subset(file_1, Reserves>0) # filtering data
head(file_1)
```

```
##      Country Year Cost.to.export Cost.of.Import Reserves
## 1   Austria 2014      1215      1150      91
## 2   Belgium 2014      1400      1240      92
## 3  Kiribati 2014       910       870     100
## 4 Mauritania 2014      1523      1640    121172
## 5 Azerbaijan 2014      3450      3460    142638
## 6 Bangladesh 2014      1515      1281    142638
##      Description      Magnitude
## 1 TOTAL RESERVES MINUS GOLD MILLIONS OF SDRS
## 2 SDR HOLDINGS MILLIONS OF SDRS
## 3 SDR HOLDING IN % OF ALLOCATION PERCENT PER ANNU
## 4 RESERVE TRANCHE POSITION MILLIONS OF SDRS
## 5 RESERVE POSITION IN THE FUND MILLIONS OF SDRS
## 6 RESERVE POSITION IN THE FUND MILLIONS OF US$
```

Let's break down the above code step-by-step to understand what each part does:

1. **Suppress Warnings:** The line `options(warn=-1)` suppresses warning messages in R. Setting `warn = -1` turns off warnings, so they don't clutter your output.
2. **Read CSV File:** The `read.csv()` command reads a CSV file from the specified path into a data frame named `file_1`.

3. Summarization Using Descriptive Statistics:

- `summary(file_1)`: Provides a summary of the data frame, including minimum, 1st quartile, median, mean, 3rd quartile, and maximum values for each column.
- `mean(file_1$Reserves)`: Calculates the mean (average) of the Reserves column.
- `median(file_1$Cost.to.export)`: Calculates the median (middle value) of the Cost.to.export column.
- `sd(file_1$Cost.of.Import)`: Calculates the standard deviation of the Cost.of.Import column, which measures the spread of values.

4. Correlation Between Variables:

The line `cor(file_1$Reserves, file_1$Cost.to.export)` calculates the Pearson correlation coefficient between the Reserves and Cost.to.export columns, indicating the strength and direction of the linear relationship between them.

5. Data Cleaning:

- **file_1 <- na.omit(file_1)**: Removes any rows in the data frame that contain missing values (NA).
- **file_1 <- subset(file_1, Reserves > 0)**: Filters the data frame to include only rows where the Reserves value is greater than 0.
- **head(file_1)**: Displays the first few rows of the cleaned data frame to check the output.

This code reads and processes the “Bank Reserves” data, providing a summary, calculating key statistics, examining correlations, and cleaning the data.

Data Manipulation

Data manipulation refers to the process of transforming and organizing raw data into a structured format suitable for analysis. It involves cleaning the data by removing errors and handling missing values, transforming data into appropriate formats, summarizing data through statistical measures, filtering out irrelevant data, and merging multiple datasets to enrich the information.

```
library(dplyr)
```

```
##  
## Attaching package: 'dplyr'
```

```
## The following objects are masked from 'package:stats':  
##  
## filter, lag
```

```
## The following objects are masked from 'package:base':  
##  
## intersect, setdiff, setequal, union
```

```
# Filtering and selecting  
data_filtered <- file_1 %>% filter(Reserves > 1000) %>% select(Country, Reserves)  
data_filtered
```

```
##           Country    Reserves  
## 1 Mauritania 1.21172e+05  
## 2 Azerbaijan 1.42638e+05  
## 3 Bangladesh 1.42638e+05  
## 4 Bahrain 2.23612e+05  
## 5 Eritrea 3.65671e+06  
## 6 Comoros 6.67121e+06  
## 7 Lesotho 3.12586e+07  
## 8 Dominica 4.81371e+07  
## 9 Armenia 6.29377e+07  
## 10 Nepal 6.37137e+07  
## 11 Fiji 6.70627e+07  
## 12 Haiti 6.89196e+07  
## 13 Myanmar 7.22678e+07  
## 14 Antigua and Barbuda 8.15597e+07  
## 15 Guinea 8.22084e+07  
## 16 Grenada 8.23371e+07  
## 17 Honduras 1.04778e+08  
## 18 Mozambique 1.08657e+08  
## 19 Afghanistan 1.28467e+08  
## 20 Liberia 1.28491e+08  
## 21 Malawi 1.36214e+08  
## 22 Maldives 1.66474e+08  
## 23 Guinea-Bissau 1.68593e+08  
## 24 Djibouti 2.41832e+08  
## 25 Ghana 2.90513e+08  
## 26 Burundi 3.22044e+08  
## 27 Arab World 3.55224e+08  
## 28 Chad 3.93382e+08  
## 29 Niger 4.18159e+08  
## 30 Cyprus 5.07894e+08  
## 31 Malta 5.32093e+08  
## 32 Argentina 5.56882e+08  
## 33 Guyana 6.31410e+08  
## 34 Luxembourg 7.30515e+08  
## 35 Mongolia 8.25720e+08  
## 36 Kyrgyz Republic 9.52977e+08  
## 37 Madagascar 1.13549e+09  
## 38 Gabon 1.27145e+09  
## 39 Burkina Faso 1.29584e+09  
## 40 Namibia 1.30825e+09
```

## 41	Moldova	1.48027e+09
## 42	Australia	1.51815e+09
## 43	Nigeria	1.51815e+09
## 44	Greece	1.55481e+09
## 45	Nicaragua	1.57308e+09
## 46	Mali	1.60448e+09
## 47	Belarus	1.67538e+09
## 48	Ethiopia	1.78087e+09
## 49	Cambodia	1.81868e+09
## 50	Ireland	1.94080e+09
## 51	Jamaica	2.08045e+09
## 52	Georgia	2.11032e+09
## 53	Mauritius	2.17877e+09
## 54	Albania	2.31391e+09
## 55	Barbados	2.37999e+09
## 56	Estonia	2.53363e+09
## 57	El Salvador	2.86882e+09
## 58	Ecuador	2.87318e+09
## 59	Dominican Republic	2.88512e+09
## 60	Equatorial Guinea	3.25194e+09
## 61	Cameroon	3.67552e+09
## 62	Iceland	3.81324e+09
## 63	Kenya	3.84896e+09
## 64	Costa Rica	4.06617e+09
## 65	Guatemala	4.96362e+09
## 66	Lithuania	6.45322e+09
## 67	Latvia	6.63180e+09
## 68	Belize	7.31808e+09
## 69	Botswana	8.70396e+09
## 70	Croatia	9.50093e+09
## 71	Finland	9.71058e+09
## 72	Bulgaria	1.09252e+10
## 73	Netherlands	1.15490e+10
## 74	Jordan	1.16893e+10
## 75	Benin	1.33851e+10
## 76	Angola	1.36438e+10
## 77	New Zealand	1.55940e+10
## 78	Colombia	1.57861e+10
## 79	Kuwait	2.02675e+10
## 80	Kazakhstan	2.07198e+10
## 81	Morocco	2.27973e+10
## 82	Canada	2.38790e+10
## 83	Chile	2.52819e+10
## 84	Lebanon	2.91029e+10
## 85	Italy	2.91961e+10
## 86	Egypt	3.22530e+10
## 87	Germany	3.85404e+10
## 88	Israel	3.86628e+10
## 89	Czech Republic	4.11566e+10
## 90	Hungary	4.40737e+10
## 91	Iraq	4.62554e+10
## 92	France	4.66335e+10
## 93	Bolivia	5.64269e+10
## 94	Mexico	6.35261e+10
## 95	Indonesia	6.35633e+10
## 96	Denmark	7.42909e+10
## 97	Bhutan	8.09865e+10
## 98	Malaysia	9.54317e+10
## 99	Libya	9.90257e+10
## 100	Algeria	1.49000e+11
## 101	Brazil	2.37000e+11
## 102	India	2.65000e+11
## 103	Euro area	2.84000e+11
## 104	Japan	6.52000e+11
## 105	Bosnia and Herzegovina	1.40860e+09
## 106	Brunei	1.24543e+08
## 107	Côte d'Ivoire	4.18159e+08
## 108	Cabo Verde	6.55544e+08
## 109	Caribbean small states	5.42946e+07
## 110	Central Europe and the Baltics	8.64034e+06
## 111	China	8.64034e+06
## 112	Congo	1.35454e+07
## 113	Dem. Rep. Congo	4.18159e+08
## 114	Early-demographic dividend	8.51171e+07
## 115	East Asia & Pacific	3.55224e+08
## 116	East Asia & Pacific (excluding high income)	5.56882e+08

## 117	East Asia & Pacific (IDA & IBRD)	6.29377e+07
## 118	Eswatini	1.51815e+09
## 119	Europe & Central Asia (excluding high income)	1.42638e+05
## 120	Europe & Central Asia (IDA & IBRD)	2.23612e+05
## 121	European Union	1.42638e+05
## 122	Fragile and conflict affected situations	2.37999e+09
## 123	Heavily indebted poor countries (HIPC)	1.67538e+09
## 124	Hong Kong SAR, China	7.31808e+09
## 125	IBRD only	1.33851e+10
## 126	IDA & IBRD total	8.09865e+10
## 127	IDA blend	1.37000e+08
## 128	IDA only	8.50000e+08
## 129	IDA total	8.50000e+08
## 130	Iran	4.83663e+09
## 131	Korea	8.66704e+08
## 132	Kosovo	5.52854e+08
## 133	Lao PDR	5.67589e+08
## 134	Late-demographic dividend	4.21000e+05
## 135	Latin America & Caribbean	3.56145e+08
## 136	Latin America & Caribbean (excluding high income)	5.52854e+08
## 137	Latin America & Caribbean (IDA & IBRD)	8.66704e+08
## 138	Least developed countries: UN classification	1.55940e+10
## 139	Low & middle income	9.94712e+09
## 140	Low income	8.54656e+08
## 141	Macedonia	1.73841e+08
## 142	Marshall Islands	1.73841e+08
## 143	Micronesia	6.52473e+03
## 144	Middle East & North Africa	9.98834e+07
## 145	Middle East & North Africa (excluding high income)	6.80996e+07
## 146	Middle East & North Africa (IDA & IBRD)	1.15490e+10
## 147	Middle income	1.81053e+10
## 148	Montenegro	4.88630e+09
## 149	Norway	8.68659e+08
## 150	OECD members	1.01866e+09
## 151	Oman	1.59694e+09
## 152	Pakistan	1.85549e+09
## 153	Palau	1.22382e+10
## 154	Panama	7.66021e+09
## 155	Papua New Guinea	1.96910e+07
## 156	Paraguay	2.17389e+10
## 157	Peru	5.64408e+09
## 158	Philippines	8.84816e+09
## 159	Poland	1.37000e+08
## 160	Portugal	8.50000e+08
## 161	Post-demographic dividend	8.50000e+08
## 162	Pre-demographic dividend	4.83663e+09
## 163	Puerto Rico	8.66704e+08
## 164	Qatar	5.52854e+08
## 165	Romania	5.67589e+08
## 166	Russia	2.75500e+05
## 167	Rwanda	9.41027e+10
## 168	SÃ£o TomÃ© and PrÃncipe	6.00264e+10
## 169	Samoa	2.85120e+09
## 170	San Marino	6.20073e+06
## 171	Sierra Leone	9.72083e+06
## 172	Singapore	4.80673e+06
## 173	Slovak Republic	4.71000e+11
## 174	Slovenia	1.55034e+10
## 175	Solomon Islands	2.38686e+09
## 176	South Africa	3.74185e+09
## 177	South Asia	2.38686e+09
## 178	South Asia (IDA & IBRD)	4.72000e+11
## 179	South Sudan	2.43046e+10
## 180	Spain	3.08627e+07
## 181	Sri Lanka	4.53000e+11
## 182	St. Kitts and Nevis	1.45195e+10
## 183	St. Lucia	1.48027e+09
## 184	St. Vincent and the Grenadines	9.44233e+08
## 185	Sub-Saharan Africa	2.27877e+06
## 186	Sub-Saharan Africa (IDA & IBRD)	1.17714e+08
## 187	Sudan	5.00000e+03
## 188	Suriname	5.00000e+03
## 189	Sweden	7.83845e+03
## 190	Switzerland	9.44233e+08
## 191	Syrian Arab Republic	3.57240e+06
## 192	Tanzania	1.89960e+05

```
## 193 Thailand 6.16654e+07
## 194 The Bahamas 2.17877e+09
## 195 The Gambia 1.38980e+09
## 196 Timor-Leste 9.99353e+07
## 197 Tonga 1.31545e+07
## 198 Trinidad and Tobago 2.06221e+07
## 199 Tunisia 1.31545e+07
## 200 Turkey 1.39420e+09
## 201 Uganda 1.56668e+08
## 202 Ukraine 1.24885e+08
## 203 United Arab Emirates 1.25780e+05
## 204 United Kingdom 1.27671e+09
## 205 United States 2.00148e+09
## 206 Upper middle income 9.68055e+07
## 207 Uruguay 1.57154e+08
## 208 Uzbekistan 2.46369e+08
## 209 Vanuatu 7.69111e+06
## 210 Venezuela 1.60448e+09
## 211 Vietnam 1.02347e+09
## 212 West Bank and Gaza 7.33807e+07
## 213 Yemen 9.89415e+06
## 214 Zambia 9.89415e+06
## 215 Zimbabwe 1.55110e+07
```

```
# Grouping and summarizing
summary_data <- data_filtered %>% group_by(Country) %>% summarise(TotalReserves = sum(Reserves))
summary_data
```

```
## # A tibble: 215 × 2
##   Country TotalReserves
##   <chr>      <dbl>
## 1 Afghanistan 128467000
## 2 Albania 23139100000
## 3 Algeria 149000000000
## 4 Angola 13643800000
## 5 Antigua and Barbuda 81559700
## 6 Arab World 355224000
## 7 Argentina 556882000
## 8 Armenia 62937700
## 9 Australia 1518150000
## 10 Azerbaijan 142638
## # i 205 more rows
```

Let's break down the above R code step-by-step:

Loading the dplyr Library:

The line `library(dplyr)` loads the dplyr library, which provides a set of functions for data manipulation in R.

Filtering and Selecting Data:

- **Data Frame:** `file_1` is your initial data frame.
- **%>% Operator:** This is the pipe operator, which allows you to chain multiple functions together.
- **filter(Reserves > 1000):** This function filters the data to include only rows where the Reserves value is greater than 1000.

- **select(Country, Reserves):** This function selects only the Country and Reserves columns from the filtered data.
- **Result:** The resulting data frame `data_filtered` contains rows where Reserves is greater than 1000, and only includes the Country and Reserves columns.

Grouping and Summarizing Data:

- **data_filtered:** Uses the filtered data frame from the previous step.
- **group_by(Country):** Groups the data by the Country column.
- **summarise(TotalReserves = sum(Reserves)):** Summarizes the data by calculating the total reserves for each country.
- **Result:** The resulting data frame `summary_data` contains the Country column and a new column `TotalReserves` that shows the sum of reserves for each country.

In summary, this code filters the dataset to include only countries with reserves greater than 1000, selects relevant columns, groups the data by country, and then calculates the total reserves for each country.

Comparative Data Analysis

The analysis provides a comparative overview of the reserves held by various countries, highlighting differences in their economic strengths and financial stability. Developed nations maintain higher reserves for stability, while developing countries face challenges due to deficits and debts. This analysis emphasizes the critical role of bank reserves in supporting trade, stabilizing currencies, and managing economic risks. The following charts and graphs showcase the comparative overview of various reserves.

Bar Chart to represent the Gold Reserves of the world

```
library(plotly)

## Loading required package: ggplot2

##
## Attaching package: 'plotly'

## The following object is masked from 'package:ggplot2':
##
## last_plot

## The following object is masked from 'package:stats':
##
## filter

## The following object is masked from 'package:graphics':
##
## layout

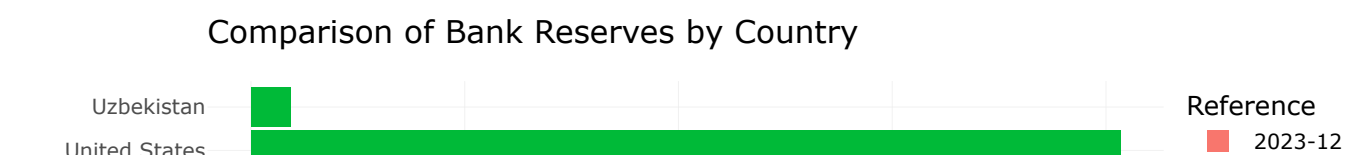
library(ggplot2)

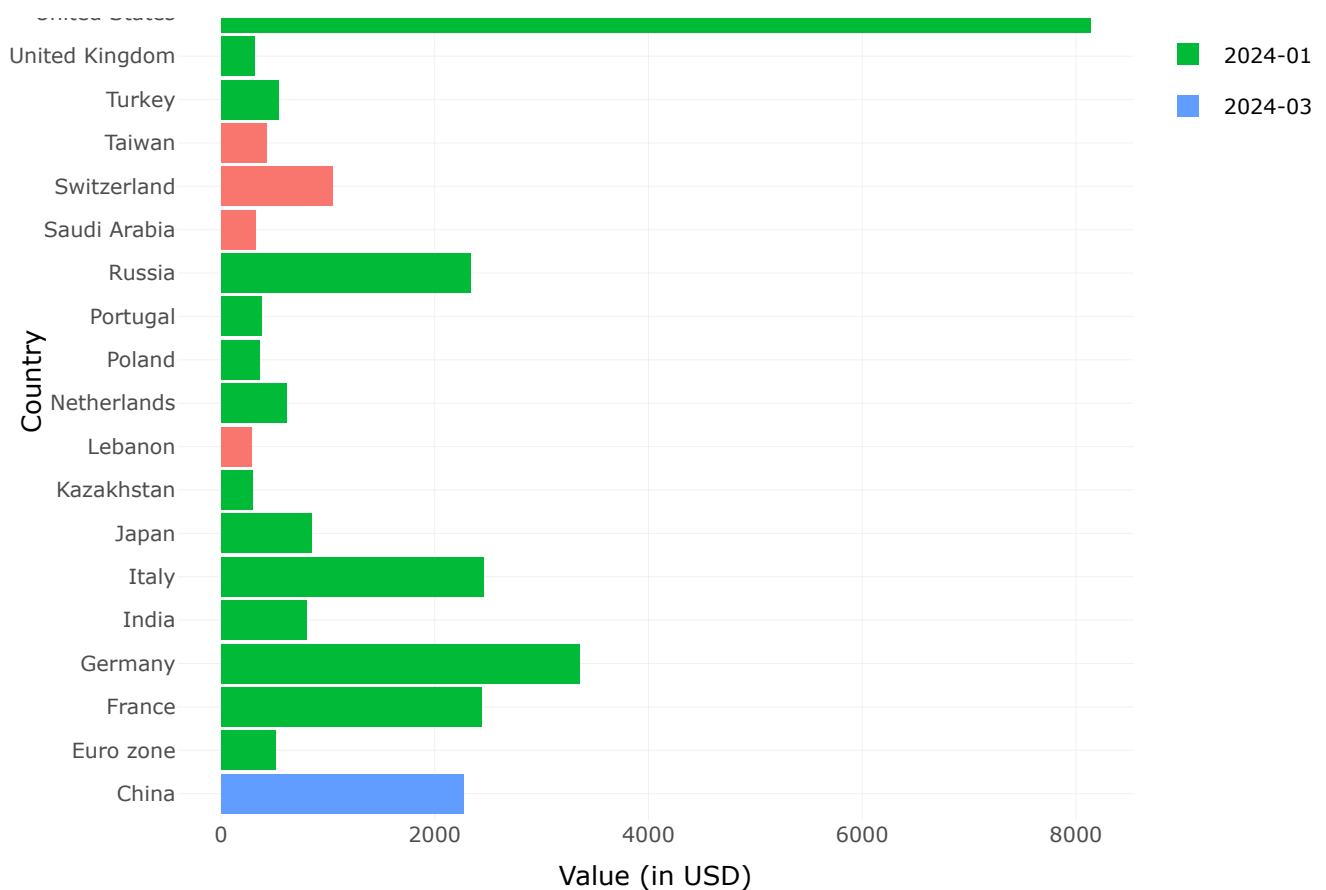
file_2 <- read.csv("C:\\Users\\Abhinav Masih\\OneDrive\\Documents\\One Drive Folder\\OneDrive\\Desktop\\RESERVES FILES\\Gold
Reserves.csv")
head(file_2)

##      Countries Reserves Reference Unit
## 1 United States    8133    2024-01 Tons
## 2      Germany    3353    2024-01 Tons
## 3        Italy    2452    2024-01 Tons
## 4        France    2437    2024-01 Tons
## 5        Russia    2333    2024-01 Tons
## 6         China    2265    2024-03 Tons

plot1 <- ggplot(head(file_2,20), aes(x = Countries, y = Reserves, fill= Reference)) +
  geom_bar(stat = "identity", position = "dodge") +
  coord_flip()+
  theme_minimal() +
  labs(title = "Comparison of Bank Reserves by Country",
       y = "Value (in USD)", x = "Country")

interactive_plot <- ggplotly(plot1)
interactive_plot
```





Let's break down the above code step-by-step:

Loading Libraries

- **plotly**: This library is used to create interactive plots.
- **ggplot2**: This library is used for creating static plots using the Grammar of Graphics.

Creating the Plot

- **ggplot(head(file_2, 20), aes(x = Countries, y = Reserves, fill = Reference))**: This initializes a ggplot object using the first 20 rows of the dataset file_2. The aes function sets the aesthetics, specifying that Countries will be on the x-axis, Reserves on the y-axis, and 'Reference' will determine the color fill of the bars.
- **geom_bar(stat = "identity", position = "dodge")**: Adds a bar plot layer to the ggplot object. The stat = "identity" argument means that the height of the bars will represent the actual values in the Reserves column. The position = "dodge" argument ensures that bars for different Description values are placed side-by-side (grouped) rather than stacked. Note: by default, it is stacked bar chart.

- **coord_flip()**: Flips the x and y axes to make a horizontal bar plot instead of a vertical one.
- **theme_minimal()**: Applies a minimal theme to the plot, which is a clean and simple design.
- **labs(title = “Comparison of Reserves by Country”, y = “Value (in USD)”, x = “Country”)**: Adds labels to the plot, including a title and axis labels.

Making the Plot Interactive

- **ggplotly** Converts the static ggplot object `plot1` into an interactive plotly object using the `ggplotly` function.

Pie Chart to compare various Income groups based on Gold Reserves

```
library(ggplot2)
library(dplyr)

# Function to classify income groups based on Reserves
classify_income <- function(d) {
  d$IncomeGroup <- ifelse(d$Reserves > 1000, "High Income",
                          ifelse(d$Reserves > 100 & d$Reserves <= 1000, "Mid Income",
                                ifelse(d$Reserves <= 100, "Low Income", NA)))
  return(d)
}

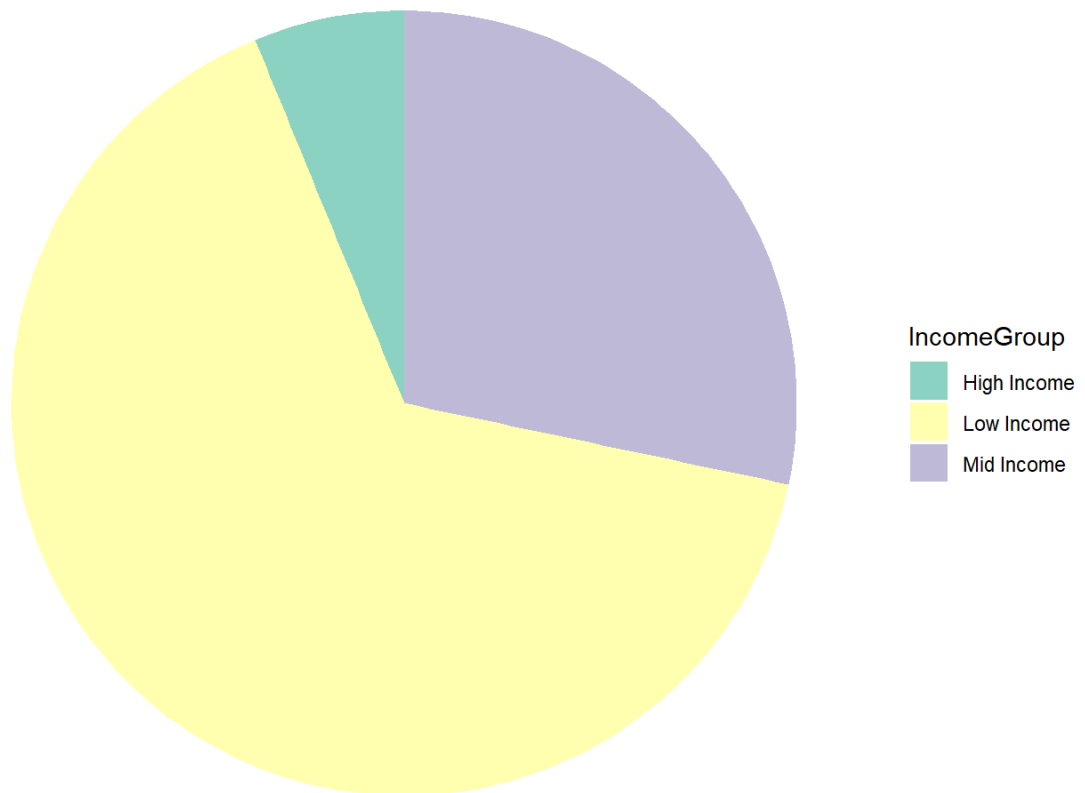
# applying to dataset file_2
file_2 <- classify_income(file_2)

# Summarize the data by income group
summary_data <- file_2 %>%
  group_by(IncomeGroup) %>%
  summarise(Count = n())

# Create a pie chart
pie_chart <- ggplot(summary_data, aes(x = "", y = Count, fill = IncomeGroup)) +
  geom_bar(width = 1, stat = "identity") +
  coord_polar("y", start = 0) +
  theme_void() +
  labs(title = "Distribution of Reserves by Income Group") +
  scale_fill_brewer(palette = "Set3")

# Display the pie chart
print(pie_chart)
```

Distribution of Reserves by Income Group



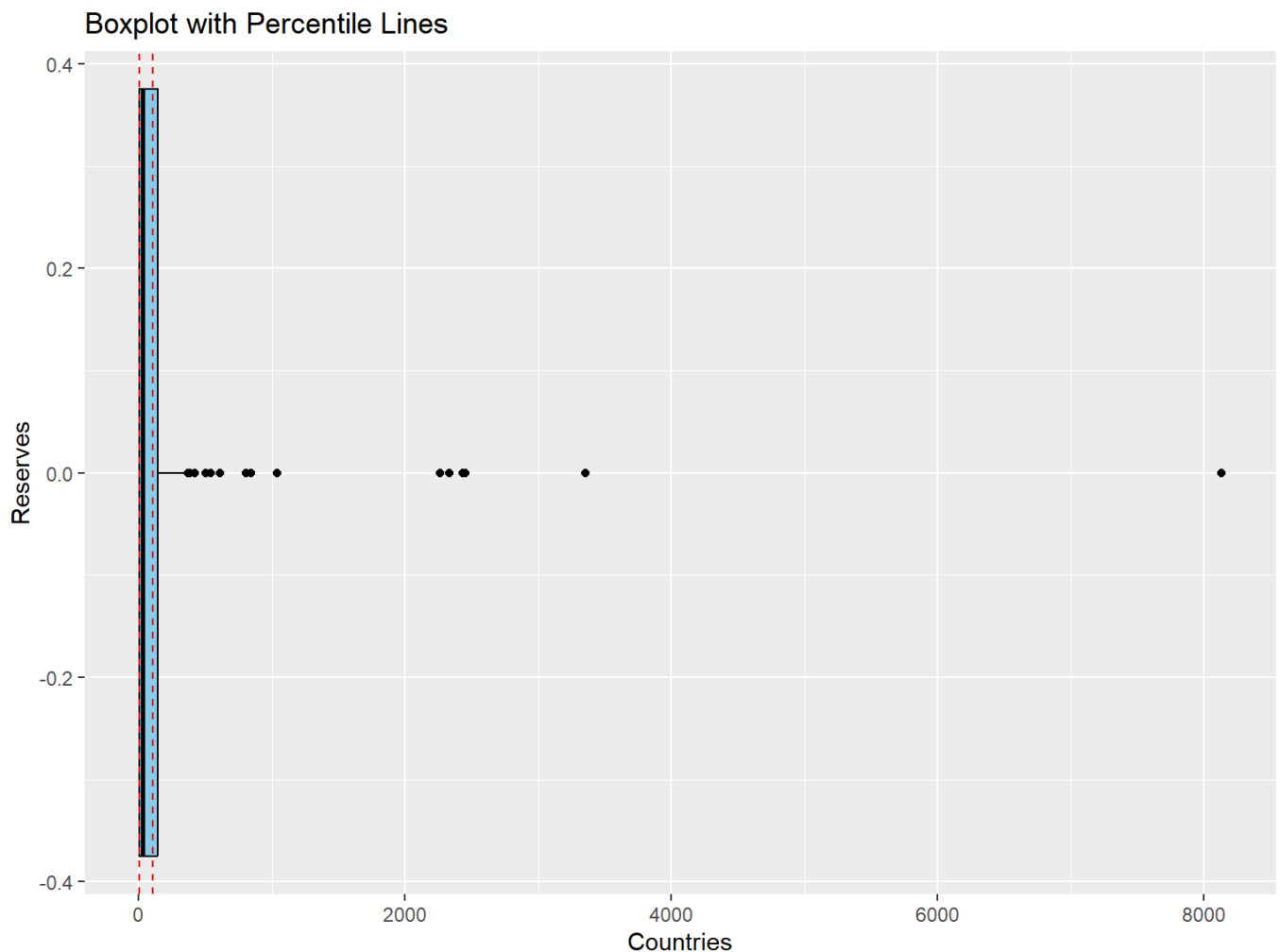
Let us go through the code step by step:

- **Classification Function:** The function classifies each row in the dataset based on the value of Reserves into High Income, Mid Income, or Low Income.
- **Apply Function:** This step applies the classification function to the dataset (`file_2`), creating a new column `IncomeGroup`.
- **Prepare Data:** The `summary_data` object groups the data by `IncomeGroup` and counts the number of countries in each group.
- **Create Pie Chart:**
 - Uses `ggplot2` to create a pie chart (`coord_polar("y", start = 0)` converts bar chart into a pie chart).
 - `geom_bar()` creates the bars for each income group, which are transformed into pie slices.
 - `theme_void()` removes axes and background for a clean look.
 - `Labs()` adds a title to the chart.
 - `scale_fill_brewer(palette = "Set3")` applies a color palette for better visual appeal.

Boxplot to compare various Income Groups

```
# To find 33rd and 67th percentiles
percentiles <- quantile(file_2$Reserves, probs = c(0.33, 0.67))

# Create a boxplot with percentile lines
ggplot(file_2, aes(x = Reserves)) +
  geom_boxplot(binwidth = 5, fill = "skyblue", color = "black")+
  geom_vline(xintercept = percentiles, linetype = "dashed", color = "red") +
  labs(title = "Boxplot with Percentile Lines", x = "Countries", y = "Reserves")
```



Let's break down the code step by step :

I. Calculate Percentiles

- **quantile(file_2\$Reserves, probs = c(0.33, 0.67))**: This function calculates the 33rd and 67th percentiles of the Reserves column in the file_2 data frame.
 - **file_2\$Reserves**: Accesses the Reserves column from the file_2 data frame.
 - **probs = c(0.33, 0.67)**: Specifies the probabilities (percentiles) to calculate, which are the 33rd and 67th percentiles.

2. Create a Boxplot with Percentile Lines

- **ggplot(file_2, aes(x = Reserves))**: Initializes a ggplot object using `file_2` as the data source and maps the `Reserves` column to the x-axis.
 - *file_2*: The data frame used for plotting.
 - *aes(x = Reserves)*: Specifies that `Reserves` should be mapped to the x-axis.
- **geom_boxplot(binwidth = 5, fill = "skyblue", color = "black")**: Adds a boxplot layer to the plot.
 - *binwidth = 5*: Adjusts the width of the boxes (though not typically used in `geom_boxplot`, it should be adjusted accordingly).
 - *fill = "skyblue"*: Sets the fill color of the boxplot to sky blue.
 - *color = "black"*: Sets the border color of the boxplot to black.
- **geom_vline(xintercept = percentiles, linetype = "dashed", color = "red")**: Adds vertical lines at the specified percentiles.
 - *xintercept = percentiles*: Positions the vertical lines at the calculated 33rd and 67th percentiles.
 - *linetype = "dashed"*: Specifies that the lines should be dashed.
 - *color = "red"*: Sets the color of the lines to red.
- **labs(title = "Boxplot with Percentile Lines", x = "Countries", y = "Reserves")**: Adds labels and a title to the plot.
 - *title*: Sets the title of the plot.
 - *x*: Labels the x-axis.
 - *y*: Labels the y-axis.

It calculates the 33rd and 67th percentiles of the `Reserves` column and creates a boxplot of the `Reserves` data. It then overlays dashed red vertical lines at the calculated percentiles and adds appropriate labels and a title to the plot.

Note: we can see that the outliers are countries like the USA, Switzerland etc having more than 1000 tonnes of Gold Reserves while most of the countries have less than 100 tonnes of Gold reserves. As evident in the pie chart, most of the countries come under low income category. Therefore, the 33rd and 67th percentiles also come out to be just 6.88 and 104.00.

Distribution Analysis

Distribution analysis examines how data points are spread within a dataset, focusing on central tendency (mean, median, mode), variability (range, variance, standard deviation), and distribution shape (symmetry, skewness, kurtosis). Key visual tools include histograms, box plots, density plots, and Q-Q plots. This analysis helps summarize data, make inferences, ensure quality control, and manage risks.

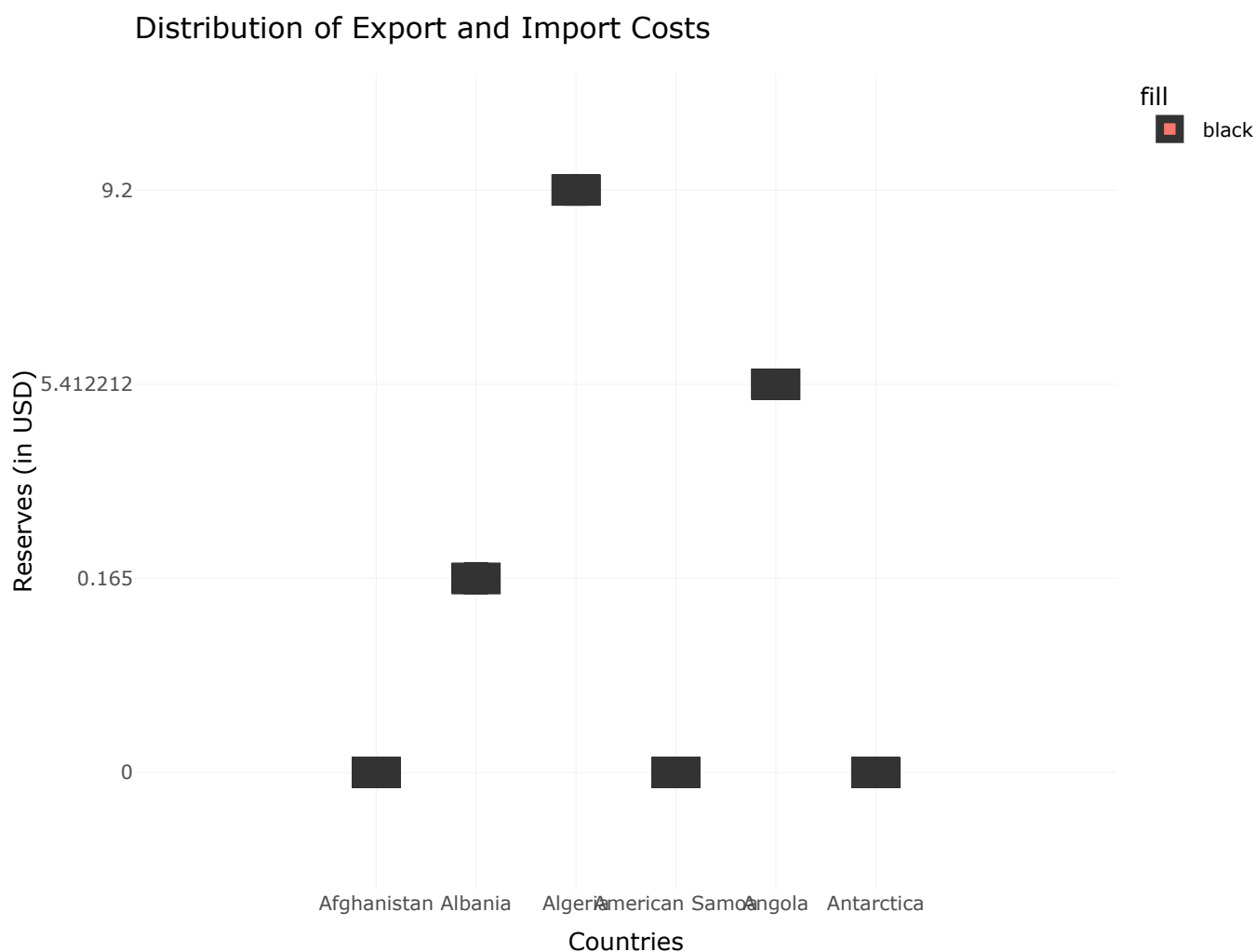
Boxplot Distribution of the Bank Reserves Data

```
library(ggplot2)

file_4<-read.csv("C:\\Users\\Abhinav Masih\\OneDrive\\Documents\\One Drive Folder\\OneDrive\\Desktop\\RESERVES FILES\\Bank
Reserves 2.csv")

plot2 <- ggplot(head(file_4), aes(x = Countries,y= Reserves, fill="black")) +
  geom_boxplot(width = 10, size = 5) +
  theme_minimal() +
  labs(title = "Distribution of Export and Import Costs",
       y = "Reserves (in USD)", x = "Countries")

interactive_plot2 <- ggplotly(plot2)
interactive_plot2
```



Let's break down the above code step-by-step to understand what each part does:

Loading Libraries

- **plotly**: This library is used to create interactive plots.
- **ggplot2**: This library is used for creating static plots using the Grammar of Graphics.

Creating the Plot

1. **ggplot(head(file_1), aes(x = Countries, y = Reserves, fill = "black")):**

- *Initializes a ggplot object using the first 6 rows of the dataset file_4 (as head defaults to 6).*
- *Sets up the aesthetic mappings (aes) where Countries is the x-axis, Reserves is the y-axis.*

2. **geom_boxplot():**

- *Adds a boxplot layer to the ggplot object to show the distribution of the export and import costs.*

3. **theme_minimal():**

- *Applies a minimal theme to the plot for a clean and simple design.*

4. **labs(title = "Distribution of Countries and Reserves", y = "Reserves (in USD)", x = "Country"):**

- *Adds labels to the plot:*
 - Title: "Distribution of Export and Import Costs"
 - y-axis label: "Reserves (in USD)"
 - x-axis label: "Countries"

5. **xlim(1, 2000):**

- *Sets the limits for the x-axis to range from 1 to 2000.*

6. **ylim(1, 2000):**

- *Sets the limits for the y-axis to range from 1 to 2000.*

Making the Plot Interactive

- **ggplotly(plot2)** Converts the static ggplot object plot2 into an interactive plotly object using the ggplotly function.

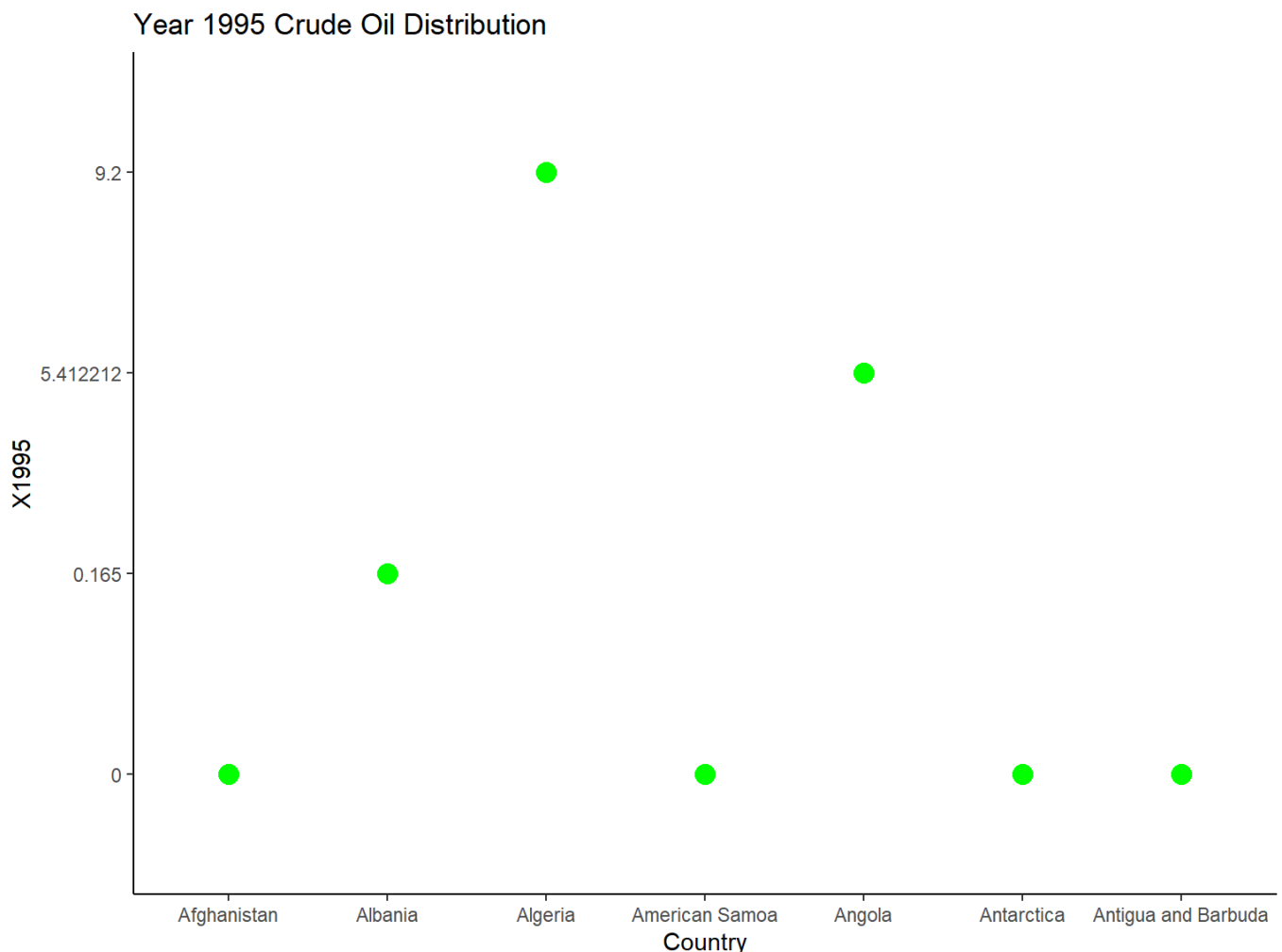
Scatter Chart distribution of Crude Oil Reserves Data

The Crude Oil Reserves Data provides comprehensive information on global crude oil reserves from 1995 to 2021. Key variables include **Countries** (Name of the country having the crude Oil Reserves), **Years** (annual oil production figures), and **exports** (annual export volumes). This dataset is valuable for energy economics research, policy-making, and understanding global oil market dynamics.

```
file_3 <- read.csv("C:\\Users\\Abhinav Masih\\OneDrive\\Documents\\One Drive Folder\\OneDrive\\Desktop\\RESERVES FILES\\Crude Oil Reserves.csv")
head(file_3)
```

	Country	X1995	X1996	X1997	X1998	X1999	X2000	X2001	X2002	X2003
## 1	Afghanistan	0	0	0	0	0	0	0	0	0
## 2	Albania	0.165	0.165	0.165	0.165	0.165	0.165	0.165	0.165	0.165
## 3	Algeria	9.2	9.2	9.2	9.2	9.2	9.2	9.2	9.2	9.2
## 4	American Samoa	0	0	0	0	0	0	0	0	0
## 5	Angola	5.412212	5.412212	5.412212	5.412	5.412	5.412	5.412	5.412	5.412
## 6	Antarctica	0	0	0	0	0	0	0	0	0
	X2004	X2005	X2006	X2007	X2008	X2009	X2010	X2011	X2012	X2013
## 1	0	0	0	0	0	0	0	0	0	0
## 2	0.165	0.165	0.19813	0.19813	0.19914	0.19914	0.19914	0.19914	0.19914	0.1724
## 3	11.314	11.8	11.35	12.27	12.2	12.2	12.2	12.2	12.2	12.2
## 4	0	0	0	0	0	0	0	0	0	0
## 5	5.412	5.412	5.412	8	9.035	9.04	9.5	9.5	9.5	10.47
## 6	0	0	0	0	0	0	0	0	0	0
	X2014	X2015	X2016	X2017	X2018	X2019	X2020	X2021		
## 1	0	0	0	0	0	0	0	0.000		
## 2	0.168332	0.168332	0.168332	0.168332	0.168332	0.168332	0.15	0.150		
## 3	12.2	12.2	12.2	12.2	12.2	12.2	12.2	12.200		
## 4	0	0	0	0	0	0	0	0.000		
## 5	9.06	9.011	8.423	8.273	9.523	8.384	8.16	7.783		
## 6	0	0	0	0	0	0	0	0.000		

```
ggplot(head(file_3,7),aes(Country,X1995))+
  geom_point(size=4, color="green")+
  labs(title= "Year 1995 Crude Oil Distribution")+
  theme_classic()
```



Let's break down the R code step-by-step to understand its functionality:

1. Reading the CSV File

- `read.csv("file_path")`: Reads a CSV file from the specified path into a data frame named `file_3`.
 - `"C:\\Users\\Abhinav Masih\\OneDrive\\Documents\\One Drive Folder\\OneDrive\\Desktop\\RESERVES FILES\\Crude Oil Reserves.csv"`: Path to the CSV file.

2. Displaying the First Few Rows of the Data Frame

- `head(file_3)`: Displays the first 6 rows of the `file_3` data frame to give you a preview of the data.

3. Creating a Scatter Plot with ggplot2

- `ggplot(data, aes(x, y))`: Initializes a ggplot object using data and maps the variables `x` and `y`.
 - `head(file_3, 10)`: Selects the first 10 rows of the `file_3` data frame.

- ***aes(Country, X1995)***: Maps the *Country* variable to the x-axis and *X1995* variable (which represents crude oil reserves in the year 1995) to the y-axis.
- **`geom_point(size=4, color="green")`**: Adds points to the scatter plot.
 - ***size=4***: Sets the size of the points to 4.
 - ***color="green"***: Sets the color of the points to green.
- **`labs(title = "Year 1995 Crude Oil Distribution")`**: Adds a title to the plot.
 - ***title***: Sets the title of the plot.
- **`theme_classic()`**: Applies a classic theme to the plot, which provides a clean, minimalistic look.

The scatter plot visualizes the crude oil reserves in the year 1995 for the first 7 countries in the dataset, with each country represented by a green point.

Temporal Relationship of Reserves

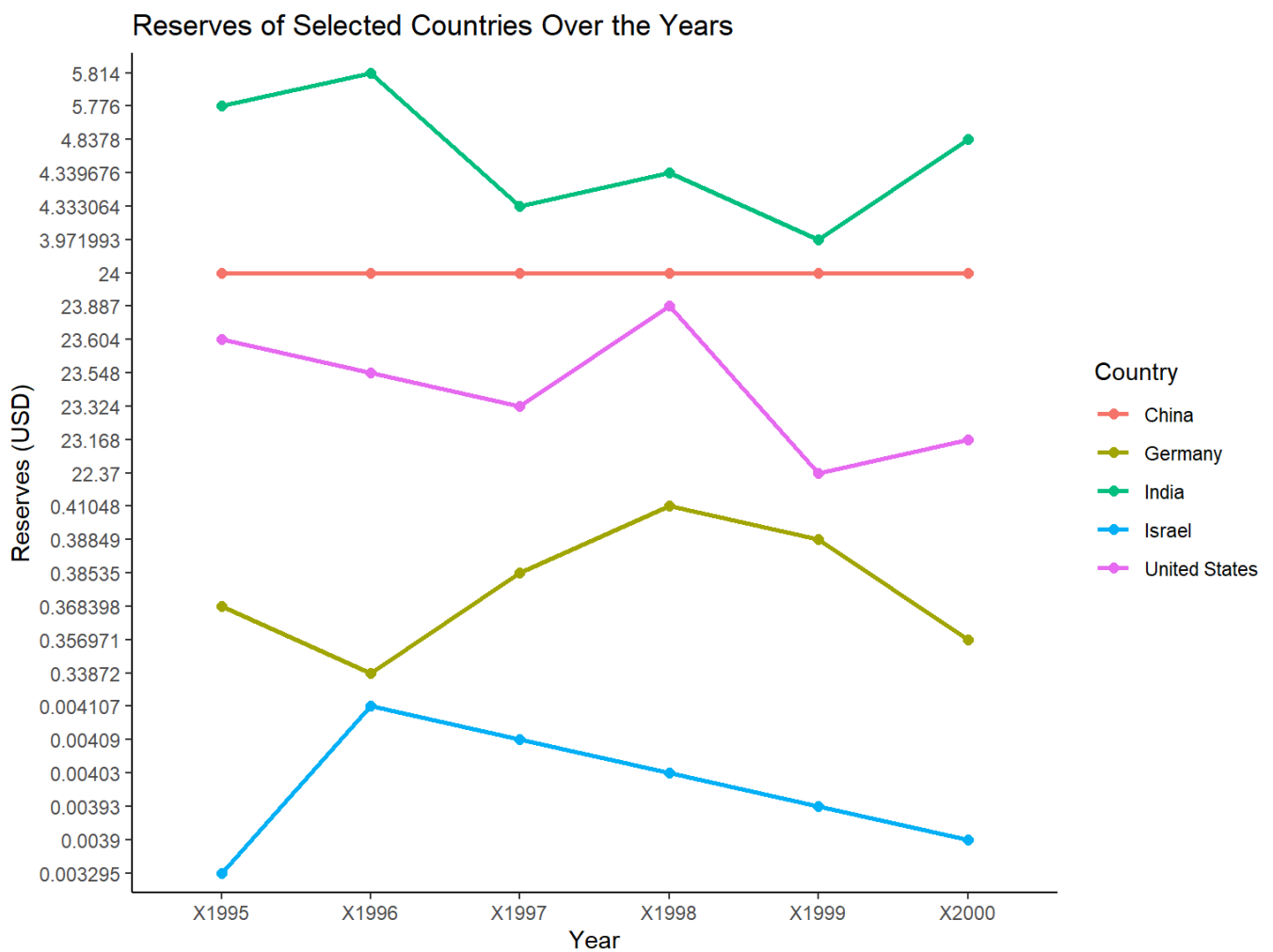
```
library(tidyr)
library(dplyr)

# reshaping the data
long_data <- file_3 %>%
  pivot_longer(cols = c("X1995", "X1996", "X1997", "X1998", "X1999", "X2000"),
               names_to = "Year",
               values_to = "Reserves")

selected_countries <- c("Germany", "China", "India", "Israel", "United States")
filtered_data <- long_data %>%
  filter(Country %in% selected_countries)

library(ggplot2)

ggplot(filtered_data, aes(x = Year, y = Reserves, color = Country, group = Country)) +
  geom_line(size = 1) +
  geom_point(size = 2) +
  labs(title = "Reserves of Selected Countries Over the Years",
       x = "Year", y = "Reserves (USD)",
       color = "Country") +
  theme_classic()
```



Correlation between Reserves and other factors

Analyzing the correlation between reserves and other factors like export costs and import costs is crucial for understanding relationships in the data.

- In order to compute Correlation we use the `cor()` function in R to calculate the correlation coefficient:

```
# Calculate correlation between reserves and export/import costs
head(file_1)
```

```
##      Country Year Cost.to.export Cost.of.Import Reserves
## 1   Austria 2014         1215         1150         91
## 2   Belgium 2014         1400         1240         92
## 3   Kiribati 2014          910          870        100
## 4 Mauritania 2014         1523         1640       121172
## 5 Azerbaijan 2014         3450         3460      142638
## 6 Bangladesh 2014         1515         1281      142638
##
##      Description      Magnitude
## 1 TOTAL RESERVES MINUS GOLD MILLIONS OF SDRS
## 2 SDR HOLDINGS MILLIONS OF SDRS
## 3 SDR HOLDING IN % OF ALLOCATION PERCENT PER ANNU
## 4 RESERVE TRANCHE POSITION MILLIONS OF SDRS
## 5 RESERVE POSITION IN THE FUND MILLIONS OF SDRS
## 6 RESERVE POSITION IN THE FUND MILLIONS OF US$
```

```
cor_res_export <- cor(file_1$Reserves, file_1$Cost.to.export, use = "complete.obs")
cor_res_import <- cor(file_1$Reserves, file_1$Cost.of.Import, use = "complete.obs")

# Print the correlation coefficients
cat("Correlation between Reserves and Export Cost:", cor_res_export, "\n")
```

```
## Correlation between Reserves and Export Cost: -0.06311038
```

```
cat("Correlation between Reserves and Import Cost:", cor_res_import, "\n")
```

```
## Correlation between Reserves and Import Cost: -0.05062525
```

- A positive correlation indicates that reserves increase as the factor increases.
- A negative correlation suggests an inverse relationship.

2. Visualize Correlation

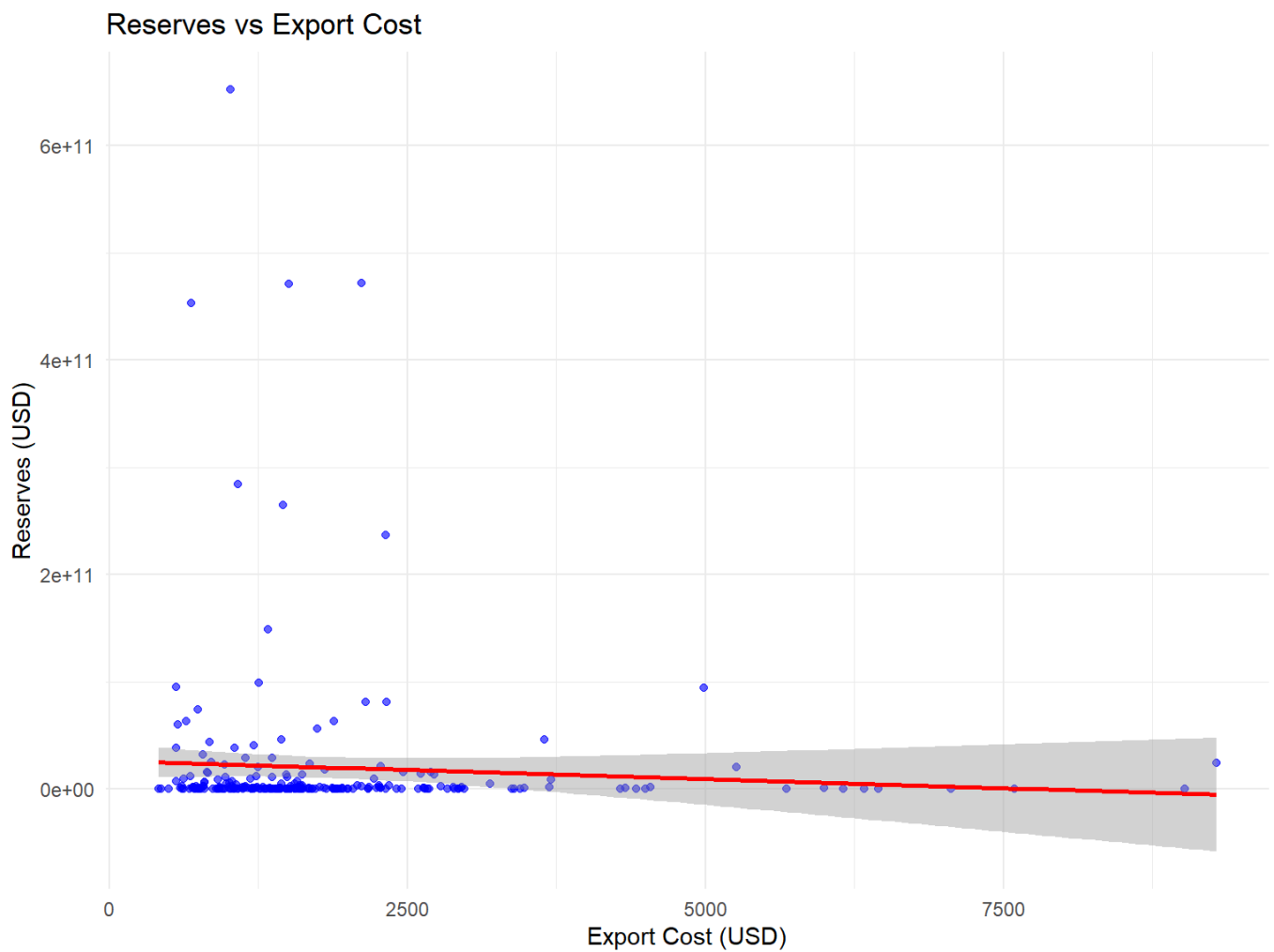
1. Scatter Plots: Scatter plots help visually inspect relationships:

```
library(ggplot2)

# Reserves vs Export Cost
ggplot(file_1, aes(x = Cost.to.export, y = Reserves)) +
  geom_point(color = "blue", alpha = 0.6) +
  geom_smooth(method = "lm", color = "red") +
  labs(title = "Reserves vs Export Cost",
```

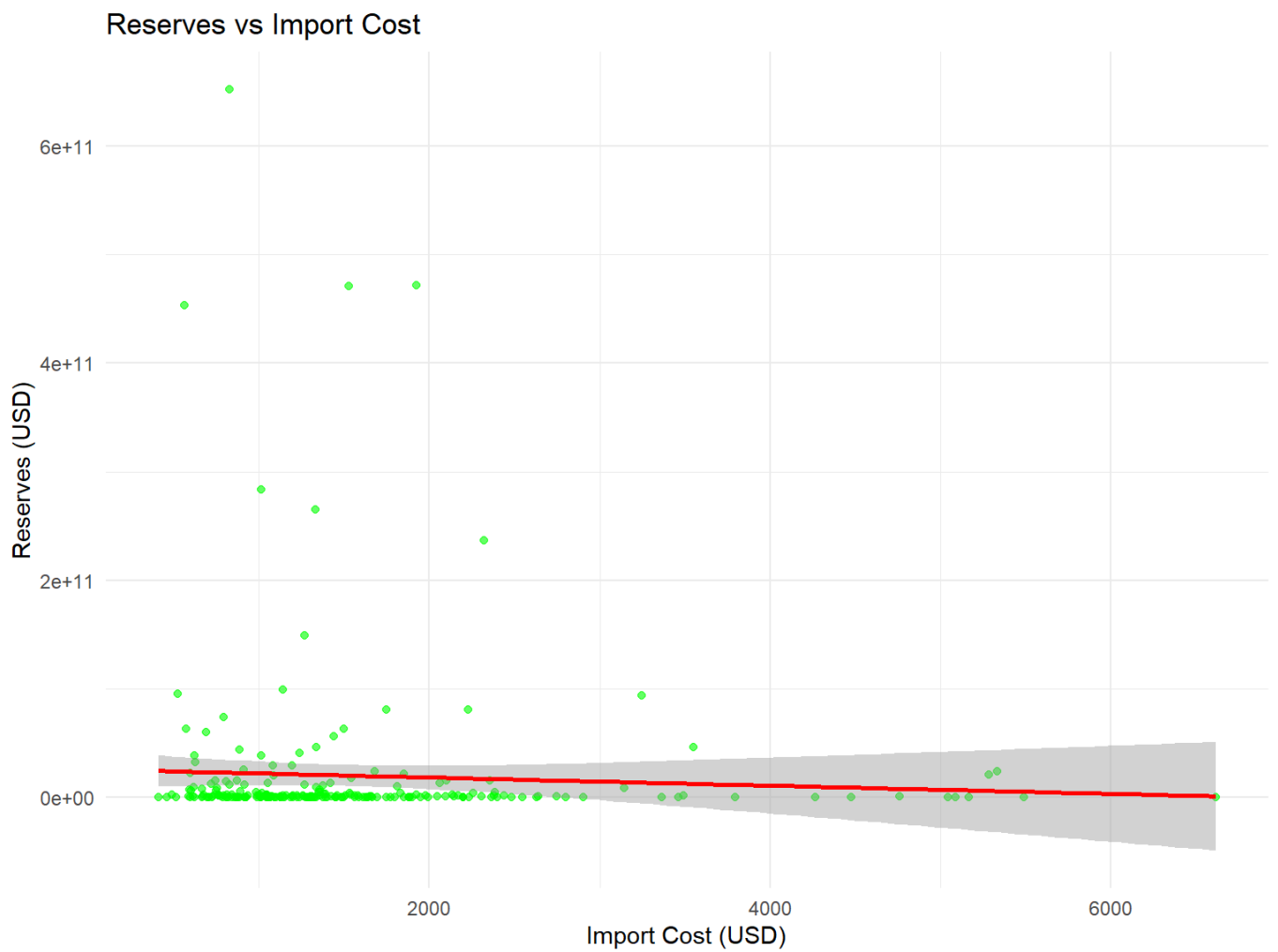
```
x = "Export Cost (USD)",
y = "Reserves (USD)" +
theme_minimal()
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



```
# Reserves vs Import Cost
ggplot(file_1, aes(x = Cost.of.Import, y = Reserves)) +
  geom_point(color = "green", alpha = 0.6) +
  geom_smooth(method = "lm", color = "red") +
  labs(title = "Reserves vs Import Cost",
       x = "Import Cost (USD)",
       y = "Reserves (USD)") +
  theme_minimal()
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



The correlation analysis and visualizations provide valuable insights into how reserves interact with export and import costs.

Identifying clusters of countries based on reserves

Clustering analysis groups countries with similar reserves, providing insights into economic similarities and disparities. The analysis helps policy makers and researchers identify regional trends, prioritize resource allocation, and develop targeted economic strategies for different clusters of countries.

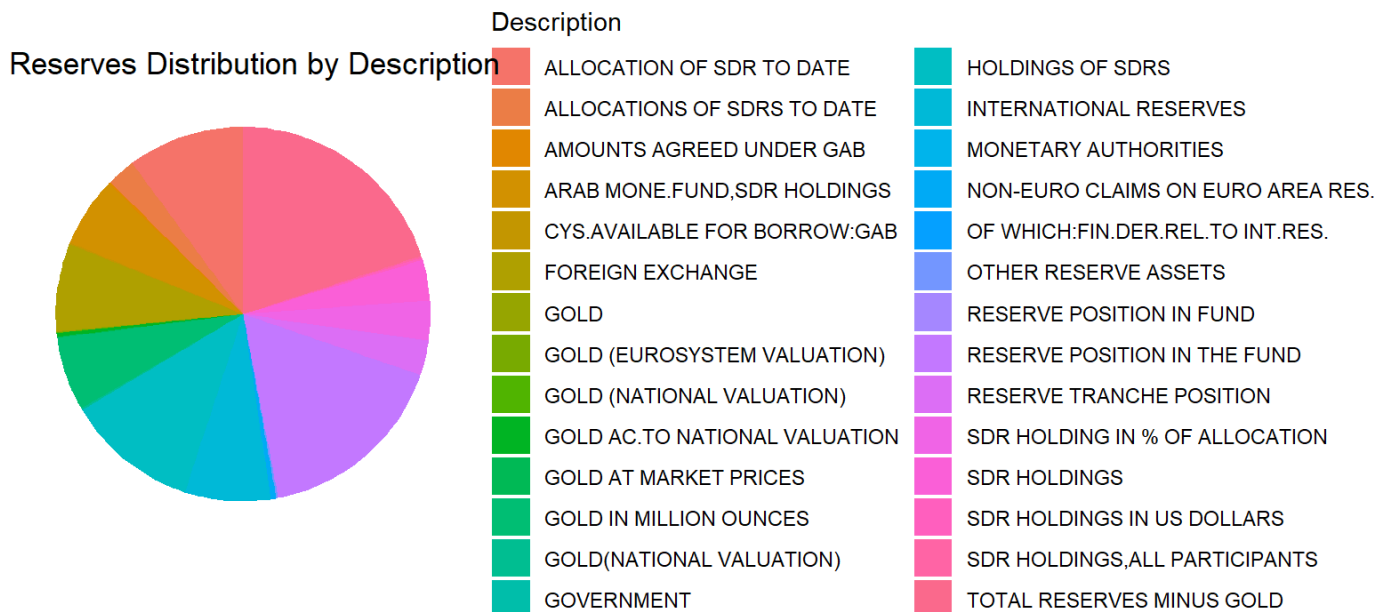
To group reserves based on the Description column and represent it as a pie chart, follow these steps:

```
# Summarize reserves by Description
library(dplyr)
data_summary <- file_1 %>%
  group_by(Description) %>%
  summarise(TotalReserves = sum(Reserves, na.rm = TRUE))
data_summary
```

```
## # A tibble: 28 × 2
##   Description                TotalReserves
##   <chr>                    <dbl>
## 1 ALLOCATION OF SDR TO DATE  471850000000
## 2 ALLOCATIONS OF SDRS TO DATE 114757526751.
## 3 AMOUNTS AGREED UNDER GAB   8034307700
## 4 ARAB MONE.FUND,SDR HOLDINGS 284000000000
## 5 CYS.AVAILABLE FOR BORROW:GAB 6529907100
## 6 FOREIGN EXCHANGE           354655906260
## 7 GOLD                        4066170000
## 8 GOLD (EUROSYSTEM VALUATION) 850000000
## 9 GOLD (NATIONAL VALUATION)   1311772000
## 10 GOLD AC.TO NATIONAL VALUATION 15323190000
## # i 18 more rows
```

```
library(ggplot2)

# Pie chart
ggplot(data_summary, aes(x = "", y = TotalReserves, fill = Description)) +
  geom_bar(stat = "identity", width = 1) +
  coord_polar("y", start = 0) +
  labs(title = "Reserves Distribution by Description",
       x = NULL, y = NULL, fill = "Description") +
  theme_void() # Removes unnecessary plot elements
```

Let's break down the R code step-by-step to understand its functionality:

1. Summarizing Reserves by Description

- **library(dplyr)**: Loads the dplyr package, which provides tools for data manipulation.
- **data_summary <- file_1 %>% group_by(Description) %>% summarise(TotalReserves = sum(Reserves, na.rm = TRUE))**:
 - *file_1 %>%*: The pipe operator %>% passes *file_1* data frame to the following functions.
 - *group_by(Description)*: Groups the data by the *Description* column.
 - *summarise(TotalReserves = sum(Reserves, na.rm = TRUE))*: Summarizes the data by calculating the total reserves (*TotalReserves*) for each description, ignoring NA values with *na.rm = TRUE*.

2. Creating a Pie Chart with ggplot2

- **library(ggplot2)**: Loads the ggplot2 package, which is used for creating plots.

- **ggplot(data_summary, aes(x = "", y = TotalReserves, fill = Description))**: Initializes a ggplot object using data_summary and maps the TotalReserves to the y-axis and Description to the fill color.
 - *x = ""*: The x-axis is set to an empty string because a pie chart doesn't use an x-axis.
- **geom_bar(stat = "identity", width = 1)**: Adds a bar plot layer.
 - *stat = "identity"*: Uses the TotalReserves values as they are.
 - *width = 1*: Sets the bar width to 1.
- **coord_polar("y", start = 0)**: Transforms the bar plot into a pie chart by using polar coordinates.
 - *coord_polar("y", start = 0)*: Positions the pie chart starting from angle 0.
- **labs(title = "Reserves Distribution by Description", x = NULL, y = NULL, fill = "Description")**: Adds labels and a title to the chart.
 - *title*: Sets the chart title.
 - *x = NULL, y = NULL*: Removes x and y-axis labels.
 - *fill = "Description"*: Sets the legend title.
- **theme_void()**: Applies a minimalistic theme by removing unnecessary plot elements like axes and grid lines.

Summary

This code summarizes Bank reserves by their description, creates a static pie chart using ggplot2, and generates an interactive pie chart using plotly. The ggplot2 pie chart provides a static visual representation.

Conclusion

In this project, we explored and analyzed the Reserves data to gain valuable insights into global distribution and trends. We undertook the following steps:

1. **Data Import and Cleaning:** We started by importing the dataset, handling missing values, and ensuring data quality through appropriate cleaning techniques.
2. **Descriptive Analysis:** We summarized the data using descriptive statistics to understand the basic characteristics of the reserves.
3. **Data Manipulation and Classification:** We differentiated countries based on their reserves, categorizing them into high, mid, and low-income groups. This classification provided a framework for further analysis.
4. **Visualization Techniques:** We utilized various visualization methods including bar plots, pie charts, and line charts to represent the data effectively. These visualizations helped in comparing the reserves across different countries and years, providing a clear picture of the trends and distributions.
5. **Percentile Analysis:** By calculating percentiles, we were able to identify specific thresholds and compare reserves distribution across countries.
6. **Interactive Visualizations:** Using tools like plotly, we created interactive charts that allowed for a more dynamic exploration of the data.

Overall, this comprehensive approach enabled us to uncover significant patterns and trends in crude oil reserves across different countries and years. The insights gained from this analysis can be valuable for policymakers, researchers, and industry stakeholders in understanding the global market dynamics and making informed decisions.

THANK YOU

THANK YOU!